

# Portfolio Optimizer (Simplified)

---

A tool that builds the "best" investment portfolio from a set of stocks — the 5th project in the series, asking a different question than the others:

- Options Pricing Engine → what should an asset be worth
  - VaR Calculator → how much could a portfolio lose
  - Stat Arb Backtester → can we exploit mispricings
  - HF Market Simulator → what actually happens when a strategy tries to trade
  - This project → what's the best possible portfolio, and how does "best" change once real limits (leverage, turnover, sector rules) are applied?
- 

## Module 1: Best Portfolio (No Limits)

---

Uses a direct formula (no trial-and-error needed) to find the portfolio that gives the best return-per-unit-of-risk (Sharpe ratio), out of 19 stocks across 5 sectors (tech, financials, energy, retail, payments).

Why it matters: this is the theoretical "ceiling" — every limit added later gives up some of this performance, and the project measures exactly how much.

Validated two ways:

1. The result must beat every single individual stock's own Sharpe ratio, since holding 100% of the best stock is technically a valid "portfolio" too.
2. Cross-checked against a completely different calculation method — both agreed exactly (Sharpe: 2.66).

Key finding (a real bug, caught and fixed): first attempt scored lower than the best individual stock — mathematically impossible if correct. Traced back to a simple mistake: the results table got sorted for neat display, but the sorted (reordered) version got used in the actual math afterward, mismatching each stock with the wrong numbers. Fixed by keeping a separate, never-reordered copy for calculations.

---

## Module 2: Best Portfolio (With Real Limits)

---

Adds three real-world rules at once: a leverage cap (can't borrow/short beyond a set amount), a turnover cap (can't trade more than X% of the portfolio at once), and sector limits (can't overload one sector).

Why it matters: no real fund can actually hold the "perfect" unlimited portfolio from Module 1 — this measures the real cost of the rules that make a portfolio actually usable.

Key finding: adding these three limits dropped the Sharpe ratio from 2.66 to 1.58 — a real, sizable cost. The turnover limit was the tightest constraint by far, not leverage or sector rules.

A false alarm, checked rather than assumed: many stocks landed at suspiciously identical weights, which can be a sign the optimizer got "stuck." Re-ran it from a totally different starting point as a check — got the exact same answer both times, confirming it was a genuine result, not a stuck search.

---

## Module 3: Is the "Best" Portfolio Actually Stable?

---

Splits the 3 years of stock data in half and finds the "best" portfolio separately on each half, to see if the answer holds up or is just noise.

Why it matters: the math behind Module 1 is exact, but it's only as good as the data feeding it — and stock data is noisy.

Key finding: one stock's ideal weight swung from 0.38 in the first half of the data to 1.77 in the second half — a huge, concrete example of how unstable these "optimal" answers can be just from which time window you happen to use.

Fix applied (partially works): a standard smoothing technique (shrinkage) reduced the instability by about 9% — a real but modest improvement, not a complete fix. Honestly reported as "helps some" rather than "solves it."

---

## Module 4: Does Any of This Actually Work When You Trade It?

---

Simulates actually trading both the constrained (Module 2) and unconstrained (Module 1) strategies once a month over 2 years, charging real trading costs each time.

Why it matters: this is the real test — does the "theoretically best" portfolio from Module 1 actually perform better once you account for real trading costs, or does its instability (found in Module 3) turn into real losses?

Key finding: the unconstrained ("perfect") strategy had to trade nearly 300% of the portfolio every single month just to keep chasing its own shifting "ideal" answer — costing 6.78% of the portfolio in fees over 2 years, versus just 1.13% for the constrained version.

Bottom line: the unconstrained strategy made more raw profit (+112% vs +62%), but the constrained strategy was actually the better strategy once risk is factored in (Sharpe 1.63 vs 1.06). The "theoretically perfect" portfolio and the "actually best" portfolio turned out to be two different things.

---

## What This Project Demonstrates

---

- Built a complete portfolio-building pipeline: best-case formula → real-world limits → stability check → real trading simulation
- Caught and fixed real bugs along the way (documented, not hidden), including:
  - A sorting bug that silently mismatched results with the wrong stocks
  - A missing stock accidentally left out when moving code from notebook to live app — caught only by comparing the live app's output against the notebook, not visible from either one alone
  - A website crash caused by a coding shortcut that confused the display tool — fixed by writing it out the normal way
- Backed every finding with a genuine check rather than trusting the first result (re-running from different starting points, splitting data in half, comparing live app vs. notebook)
- Deployed as a live, interactive Streamlit web app with adjustable limits and parameters